EY

Enterprise Data Center Storage Architecture

NAS · SAN · Fibre Channel · SAN Switches

Presented by: Navdeep | Audience: Arun | In-Depth Technical Session

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Enterprise Data Center

Storage Architecture

Understanding NAS, SAN, Fibre Channel & SAN Switches

Presented by: Navdeep

Session for: Arun

NAS

SAN

Fibre Channel

SAN Fabric

Agenda

What We Will Cover Today

01 **Data Center Storage Fundamentals**
Why enterprises need dedicated storage

02 **Storage Architecture Types**
DAS, NAS, SAN — differences explained

03 **NAS Deep Dive**
File-level storage, protocols, use cases

04 **SAN Deep Dive**
Block-level storage, components, architecture

05 **Fibre Channel Networking**
FC protocol, speeds, and components

06 **SAN Switches & Fabric**
Zoning, multipathing, HA design

07 **Enterprise Design & Best Practices**
Real-world architecture and business value

Agenda

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01 **Data Center Storage Fundamentals**
What storage is and why it matters

02 **Storage Architecture Types**
DAS, NAS, and SAN overview

03 **NAS Deep Dive**
File-level storage, protocols, use cases

04 **SAN Deep Dive**
Block-level storage, components, LUNs

05 **Fibre Channel & SAN Switches**
FC protocol, fabric, zoning, multipathing

06 **Comparison & Enterprise Design**
NAS vs SAN, real-world architecture

Session Topics

Why Enterprises Need Dedicated Storage

The Case for Centralized Storage Infrastructure

Data Survival

Data must persist even if a server crashes. Local disks die with the server.

Shared Access

Multiple servers (web, app, DB) must read/write the same data simultaneously.

Performance

Enterprise apps demand high IOPS, low latency, and consistent throughput.

Scalability

Storage grows independently of compute — add capacity without server changes.

Security & Compliance

Centralized access control, encryption, and auditing meet regulatory standards.

Disaster Recovery

Replication to remote sites ensures business continuity under any failure.

Storage Architecture Types

Three Models — Different Needs, Different Solutions

DAS

Direct Attached Storage

- Directly connected to ONE server
- Uses SATA / SAS interfaces
- Fastest for single-server use
- Cannot be shared
- No networking overhead
- Limited by server capacity

USE CASE

OS disks, test/dev, small apps

NAS

Network Attached Storage

- Shared file storage over Ethernet
- Protocols: NFS, SMB/CIFS
- Multiple clients share same folders
- File-level access
- Easy to deploy and manage
- Moderate performance

USE CASE

File servers, logs, media, home dirs

SAN

Storage Area Network

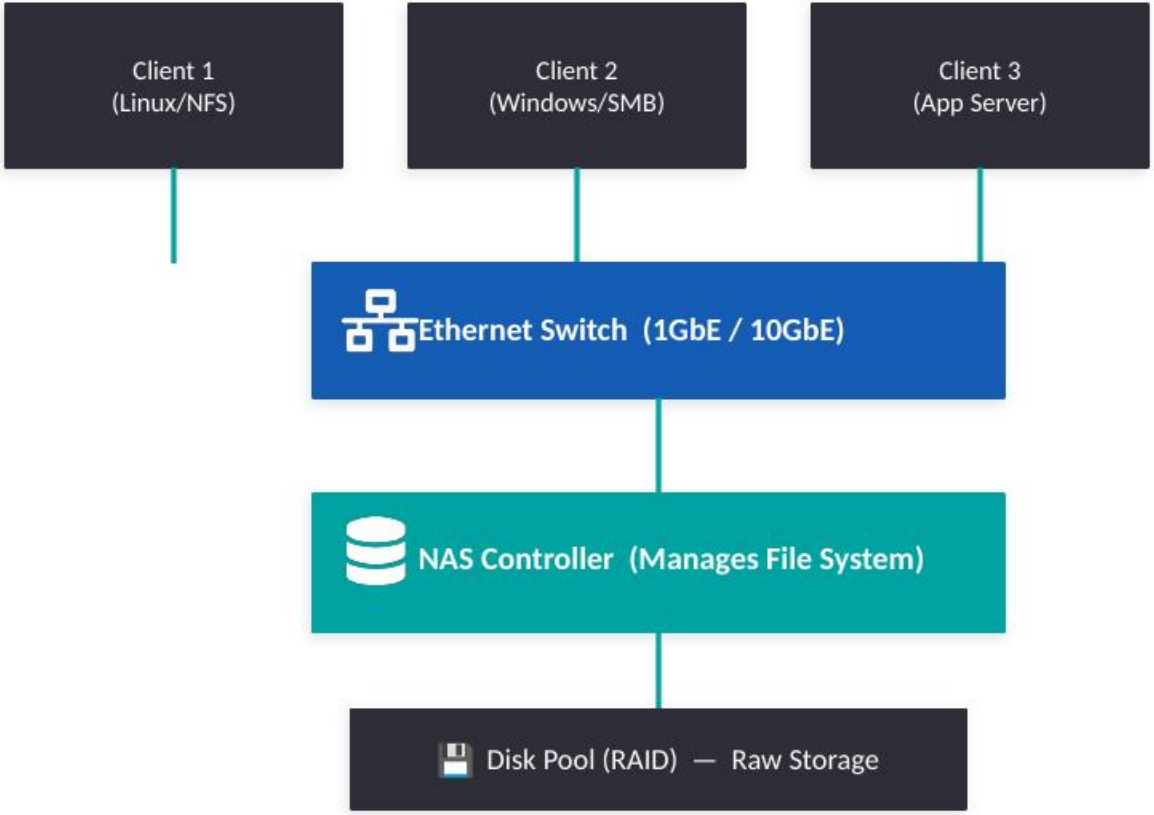
- Dedicated high-speed storage network
- Block-level access (raw LUNs)
- Appears as local disk to server
- Fibre Channel or iSCSI
- Highest performance
- Enterprise-grade HA

USE CASE

Databases, VMs, critical workloads

NAS — Network Attached Storage Architecture

File-Level Storage Over Ethernet



KEY CONCEPTS

Protocol:

NFS (Linux) or SMB/CIFS (Windows) over TCP/IP

Access Type:

File-level — clients see folders and files

Shared:
Multiple clients access the same filesystem simultaneously

FS Ownership:

NAS controller owns and manages the file system

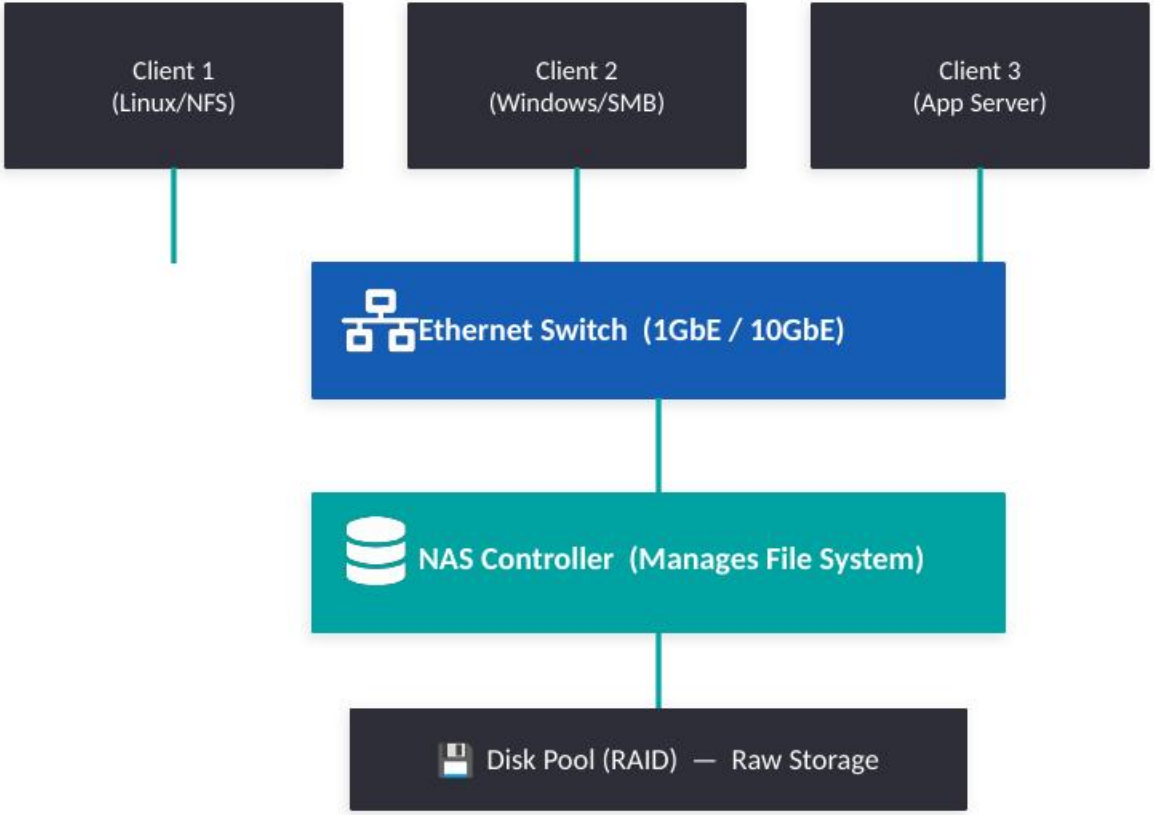
Latency:
Slightly higher than SAN due to file metadata overhead

Deployment:

Easy — plug into Ethernet, configure shares, done

NAS — Network Attached Storage Architecture

File-Level Storage Over Ethernet



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Storage Architecture Types

01 DAS

Direct Attached Storage

- Directly attached to ONE server
- SATA / SAS disks
- No network involved
- Not shareable

Use Case: OS disks, test environments

02 NAS

Network Attached Storage

- File-level storage over Ethernet
- Appears as shared folder
- Protocols: NFS, SMB/CIFS
- Easy to manage

Use Case: File servers, home directories

03 SAN

Storage Area Network

- Block-level storage
- Appears as local disk
- Dedicated high-speed network
- Max performance

Use Case: Databases, VMs, mission-critical

SAN — Storage Area Network Architecture

Block-Level Storage Over a Dedicated Fabric

Servers present storage as a LOCAL DISK — no awareness of network

NAS – Network Attached Storage

NAS provides file-level storage over a standard Ethernet network. It appears as a shared folder — accessible by any authorized user or server.

Protocols

- NFS (Network File System)

Used by Linux / Unix servers

- SMB / CIFS (Server Message Block)

Used by Windows servers and PCs

Use Cases

- Centralized file servers & shared drives
- User home directories
- Application logs and media storage
- Collaboration environments

Block Storage — How LUNs Work

The Foundation of SAN Performance

SERVER PERSPECTIVE

- Application (Oracle, SQL Server)
- File System (NTFS / EXT4 / XFS)
- Volume Manager / OS
- HBA Driver
- LUN (Looks like local disk /dev/sdb)

FC / iSCSI

STORAGE ARRAY

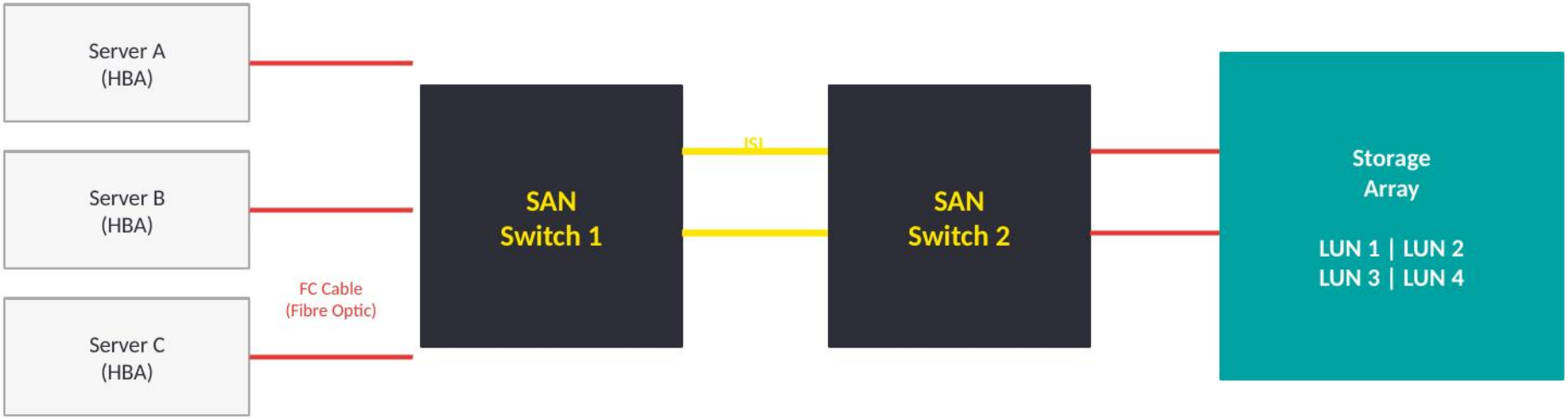
- Storage Controller
- LUN-1 (DB) — 2TB
- LUN-2 (VMs) — 5TB
- LUN-3 (Apps)— 1TB
- Physical Disks (SSD / SAS / NL-SAS)

WHY BLOCK STORAGE?

- OS controls FS — maximum flexibility
- Lowest latency possible
- ACID-compliant DB storage
- Consistent block I/O
- Supports clustering (RAC, MSCS)

SAN – Storage Area Network

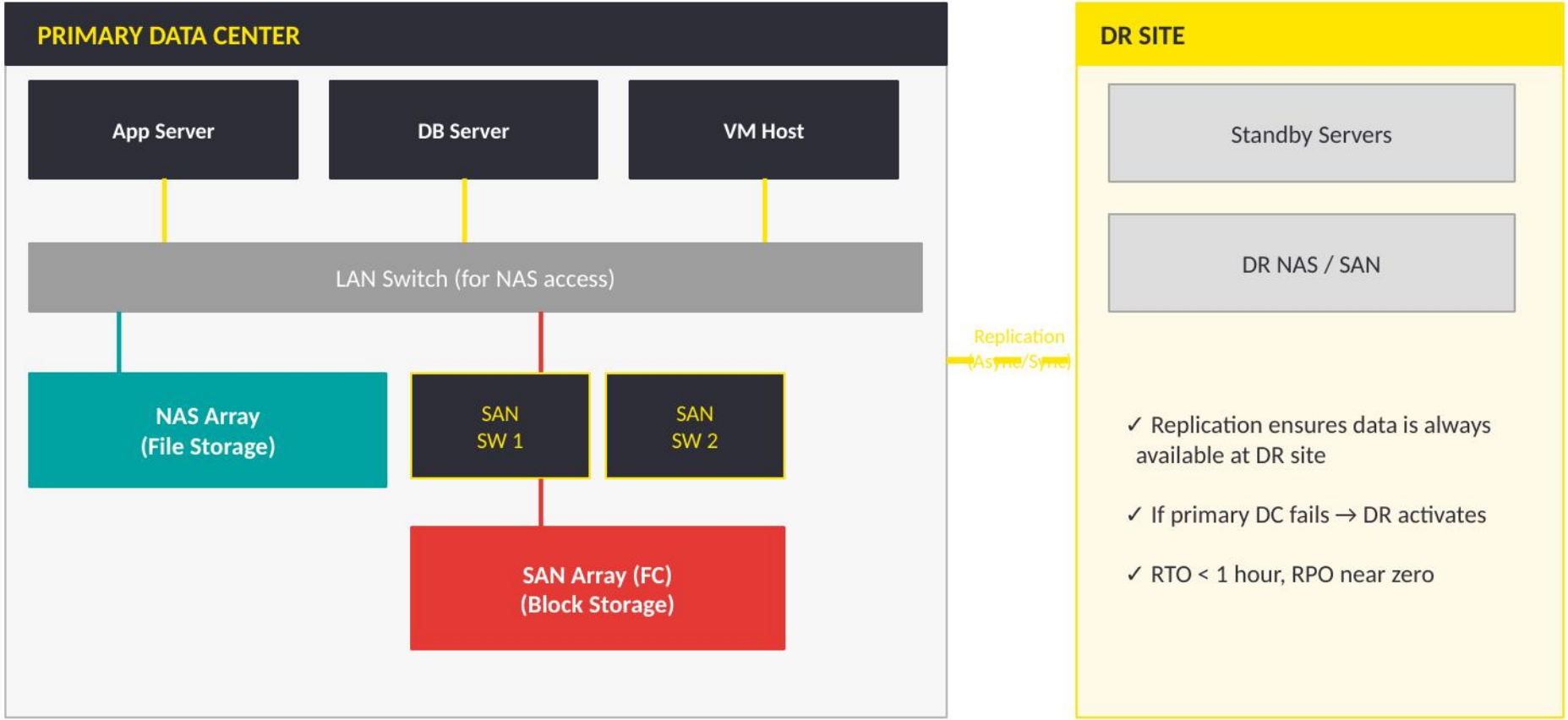
SAN provides block-level storage over a dedicated high-speed network. The server sees storage as a raw disk (LUN) — it formats and manages the filesystem itself, just like a local drive.



Key SAN Components

HBA	Host Bus Adapter — NIC equivalent for Fibre Channel in servers
LUN	Logical Unit Number — a raw block volume presented to a server
SAN Switch	Dedicated fabric switch routing FC traffic between servers & storage
ISL	Inter-Switch Link — connects two SAN switches for fabric redundancy

NAS vs SAN – Full Comparison

	NAS	SAN
Storage Type	File-level	Block-level
Network	Ethernet (IP)	Fibre Channel / iSCSI
Protocols	NFS, SMB/CIFS	FC-SCSI, iSCSI
Performance	Moderate	Very High
Who Manages FS	NAS device (built-in)	Server (OS level)
Scalability	Easy — add more capacity	Highly scalable
Cost	Lower — uses existing LAN	Higher — dedicated fabric
Use Cases	File sharing, user data, logs	DBs, VMs, mission-critical
Complexity	Simple to deploy & manage	Requires expertise

Typical Enterprise Data Center Storage Design